

Smart Surveillance for Coconut Tree Health: YOLOv11 Nano-Based Detection of Red Palm Weevil

Sarveshwaran B

Department of Electronics and Communication Engineering
Amrita School of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India.

ch.en.u4ece22041@ch.students.amrita.edu

M. Diviya

School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India.

diviya.m@vit.ac.in

JayanthRagavan M

Department of Computer Science Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Chennai, India.

ch.en.u4cse22121@ch.students.amrita.edu

K. Ashwini

Department of Computer Science Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Chennai, India.

k_ashwini@ch.amrita.edu

Abstract— The red palm weevil (*Rhynchophorus ferrugineus*) is a major pest of coconut palms, with significant agricultural losses if not treated early. Hand inspection is time-consuming and less effective, hence the use of automated detection techniques. In this paper, a deep learning-based technique that employs the YOLOv11 Nano algorithm for real-time detection of the red palm weevil in coconut palms is presented. YOLOv11 Nano, the light and lightweight object detection model, is implemented to achieve detection accuracy at minimum computational cost and thus can be implemented on the edge device. The model is trained on red palm weevil image data gathered under various environment conditions for maximizing robustness. Performance is tested through major measures like precision, recall, F1-score, accuracy, and mAP. The model achieved a precision of 93.42%, recall of 95.95%, F1-score of 94.67%, accuracy of 95.95%, and mAP@0.5 of 54%, demonstrating strong and balanced detection capabilities. Experimental results prove the model's efficacy in accurately detecting red palm weevil infestations and providing a possible solution for precision agriculture and pest control.

Keywords— Deep Learning, Object Detection, YOLOv11 Nano, Red Palm Weevil, Coconut Tree, Precision Agriculture.

I. INTRODUCTION

The red palm weevil (RPW) (*Rhynchophorus ferrugineus*) is one of the most devastating pests of coconut and palm plantations worldwide. Originally prevalent in Southeast Asia, the infestation organism has now dispersed in Africa, the Middle East, and Southern Europe, causing stupendous economic loss. It is estimated by the Food and Agriculture Organization (FAO) that red palm weevil infestations cause up to \$500 million in yearly damages worldwide to millions of date and coconut palms. Infestations start as female weevils deposit eggs within the soft tissues of the palm tree, where larvae hatch and tunnel inside the trunk of the palm, consuming internal tissue. Because these infestations are not observable for several weeks, farmer's visual identification is usually late, causing irreparable loss. Traditional methods for managing Red Palm Weevil (RPW) infestations rely on manual inspections and pheromone traps, but these approaches have significant drawbacks. They are time-consuming and labor-intensive, requiring frequent

professional monitoring, which is impractical for large plantations. Moreover, early-stage infestations often show no visible signs, resulting in low detection accuracy. By the time symptoms appear, significant internal damage may have already occurred. This delayed diagnosis leads to the removal of severely infected trees, causing substantial losses for farmers. Advancements in computer vision and deep learning have revolutionized pest detection in agriculture by overcoming the limitations of traditional image processing, which relied on manually crafted features. Convolutional Neural Networks (CNN) can automatically learn and extract important patterns from large image datasets. Among object detection algorithms like Faster R-CNN, SSD, and YOLO, the YOLO models stand out for their speed and efficiency, making them particularly well-suited for real-time agricultural applications. In this work, YOLOv11, the latest improvement in the YOLO series of object detection models. Compared to its earlier counterparts, YOLOv11 offers startling improvements in three significant areas: model size, inference speed, and detection accuracy. It is optimized to run faster, and it is thus perfectly fit for real-time use, especially in environments where resources are scarce. The model is also lightweight, consuming less computational capacity without compromising on performance. Above all, it boasts a high detection rate of 95%, and that makes it highly dependable for critical detection applications. Following these strengths, proposed research work intends to employ YOLOv11 for the detection of Red Palm Weevil, a devastating pest infesting coconut and palm trees. Based on images from a field installation, the model can identify the weevil infestation at its initial stage such that farmers and plant disease experts can respond quickly and prevent extensive damage. This precision, speed, and efficiency put together make YOLOv11 an ideal choice for this real-world agricultural issue. The system addresses key challenges in pest detection by training on a dataset of over 10,000 labeled RPW images collected from real coconut plantations and research sources. It is designed to differentiate RPWs from similar insects, tree textures, and background noise. The model is deployed on low-power IoT devices like drones and smart farm systems for real-time monitoring. Its performance is evaluated using metrics such as Mean Average Precision (mAP), precision-recall curves, and inference speed to ensure both accuracy and efficiency. The below research work depicts the existing work.

II. LITERATURE SURVEY

Murcia Labaña et al. [1] in 2018 developed a mobile application prototype of PestDetect from the use of CNN and

mobile camera technology for real-time pest and disease detection in agriculture to enable early diagnosis and intervention Koubaa et al. [2] in 2020 developed an IoT-based system for detection of Red Palm Weevil (RPW) in palm trees using embedded sensors and cloud monitoring with early warning generation and reducing dependence on human inspection. Also in 2020, Shakir Sumit et al. [3] compared YOLO with Mask R-CNN and concluded that YOLO is faster and more accurate, particularly in the case of complex scenes, and therefore more appropriate for real-time pest detection systems. Singh et al. [4] developed an accurate and real-time red palm weevil and stem bleeding disease and insect detection system for coconut trees in 2021 using InceptionResNetV2 and MobileNet. Kagan, Fuhrmann Alpert, and Fire [5] suggested a large RPW surveillance model based on deep learning using aerial and street-view images to automatically label palms and detect infestation. Hazem Alaa Eldin et al. [6] suggested a hybrid approach that combined thermal and hyperspectral imaging with deep learning for improving RPW detection efficiency and accuracy under various field conditions. Turkoglu et al. [7] proposed PlantDiseaseNet in 2022, a deep architecture based on ensemble consisting of multiple CNN and SVM classifier, which was superior to individual models for plant disease and pest classification. Alsanea et al. [8] proposed a real-time system for RPW localization based on R-CNN that outperformed bounding box-based methods in terms of pest detection accuracy. Alibabaei et al. [9] in a review, portrayed the promise of deep learning to agriculture and outlined harsh challenges such as poor generalizability, climatic uncertainty, and lacking quality labeled data, calling for further work. In 2023, Boulila et al. [10] presented a new approach of Red Palm Weevil early detection through the conversion of insect sound patterns into images, which were processed using deep learning algorithms like ResNet50V2, VGG19, and DenseNet121 based on the optimal positioning of sensors. Kumar et al. [11] enhanced insect detection by presenting the use of YOLOv5 architecture with spatial and channel attention modules, leading to more accuracy in real-time and limited-resource environments. Alif and Hussain [12] conducted an in-depth analysis of YOLO iterations (v1–v10) and their development, as well as widespread applications across agriculture, from pest identification to livestock monitoring. Jha et al. [13] in 2023 showed real-time herbal plant identification with YOLOv5 on NVIDIA Jetson Nano for field identification efficiency and low power consumption. Zhang et al. (2025) [14] introduced an aphid detection model for *Lycium barbarum* based on YOLOv11 with LAE attention and Shape-IoU Loss for higher accuracy at the early detection stage. Wang et al. (2025) [15] proposed SMC-YOLO, a faster YOLO model for maize pest detection, and added SPCPM and MDFEM modules to improve classification accuracy and inter-class generalization. The below table 1 summarizes the above literature surveys. AI-driven methods using deep learning, UAV, and multispectral imaging enable early detection of red palm weevil (RPW) infestations in coconut palms. CNN analyze RGB, thermal, and multispectral data, while U-Net and Mask R-CNN support real-time UAV monitoring. Acoustic sensing detects hidden larvae beyond visible signs. These scalable, automated approaches surpass manual inspections by fusing multimodal data for greater precision and speed. Recent studies combine satellite imagery, UAV imaging, acoustic sensing, and IoT data with deep learning (CNN, DAS) to

identify RPW before visible damage. Big data and geospatial fusion enable large-scale, real-time monitoring, promoting sustainable pest management. Recent studies depict significant advancements in real-time pest detection using light-weight and multimodal approaches, such as YOLOv11 in detecting small pests in dense fields and image, infrared, and sensor fusion models for more robust and accurate detection. While these methods hold the promise of achieving success in scalable, efficient, and useful farm use, they also fall short of each other in the same ways: dependency on good-quality data, weather sensitivity, computational cost, poor generalizability to crops and regions, and inability to be easily deployed in real time. This indicates the need to have light, agile, and scalable solutions that are interpretable and optimized with respect to both accuracy and real-world usability.

Table 1: Summary of the Literature Survey

S no	Author & Year	Model	Dataset	Accuracy	Trade-offs
1	Murcia Labaña et al., 2018 [1]	CNN-based mobile app (PestDetect)	Smartphone images	>90%	Real-time pest/disease detection, mobile-focused
2	Koubaa et al., 2020 [2]	IoT + Cloud monitoring	Embedded sensors on palm trees	Not Specified	Remote palm health monitoring, reduces manual inspection
3	Sumit et al., 2020 [3]	YOLO vs Mask R-CNN	General object detection	YOLO $\approx$ 90%, Mask R-CNN $\approx$ 85%	YOLO faster & more accurate for real-time pest detection
4	Singh et al., 2021 [4]	MobileNet + InceptionResNetV2	Coconut tree diseases/pests	96.94%	Real-time detection with efficient models
5	Kagan et al., 2021 [5]	Deep learning on aerial & street imagery	100,000+ images	Not specified	Large-scale automatic mapping, scalable pest surveillance
6	Alaa Eldin et al., 2021 [6]	Thermal + hyperspectral imaging + DL	Field environments	Not specified	Hybrid imaging for improved accuracy and speed
7	Turkoglu et al., 2022 [7]	Ensemble CNN + SVM (PlantDiseaseNet)	Plant diseases and pests	97.56% (majority vote), 96.83% (fusion)	Hybrid models improve classification
8	Alsanea et al., 2022 [8]	Real-time pest localization using R-CNN	RPW detection	99%	Combines CNN and bounding box detection for targeted control

9	Alibaba et al., 2022 [9]	Review of DL in agriculture	Various agricultural tasks	Not specified	Challenges: generalizability, dataset scarcity, environmental variability
10	Boulila et al., 2023 [10]	Sound pattern analysis + DL (ResNet50V2, etc.)	Acoustic signals of RPW	Not specified	Novel approach using insect sound for early detection
11	Kumar et al., 2023 [11]	YOLOv5 + attention modules	Insect detection	93% (stated improvement)	Lightweight, suitable for real-time and resource-limited devices
12	Alif & Hussain, 2023 [12]	Review of YOLO versions	Agriculture applications	Not specified	Evolution of YOLO models in speed, accuracy, and applications
13	Jha et al., 2023 [13]	YOLOv5 on Jetson Nano	Herbal plant detection	90–92%	Energy-efficient real-time deployment on edge devices
14	Zhang et al., 2025 [14]	YOLOv11 + LAE attention + Shape-IoU loss	Aphid detection in Lycium	95%	Improved bounding box quality and accuracy
15	Wang et al., 2025 [15]	SMC-YOLO (refined YOLO)	Maize crop pest detection	94–96%	Enhanced feature modules, better generalization

III. EXPERIMENTAL SETUP

A. Dataset and Annotation

The dataset used [16] in this study contains 1,485 labeled images, divided into training (70%, 1,039 images), validation (20%, 297 images), and test (10%, 149 images) sets, following standard practices. Annotations follow the YOLO format, which specifies object class, center coordinates, and normalized width and height for bounding boxes. Labeling was performed using the open-source tool Labellmg. Before training, the dataset undergoes several preprocessing steps and techniques to enhance model performance, as detailed in the following sections.

B. Dataset Preprocessing

1. Pre-processing Considerations

The data employed in the present research was already pre-processed adequately, so no further standard preprocessing techniques were necessary before training. Thus, model training was started directly with raw annotated images. Nonetheless, an unforeseen issue arose in testing: the model

showed a bias towards identifying black-colored areas as insects. This was attributed to the overwhelming figure of insects having black-colored stings in the training images. Figure 1 represents the augmented images of insects. As a result, the model would relate the color black to the presence of insects, independent of true shape or texture. This effect is due to the model "shortcut learning" by learning mostly color features, instead of learning more stable representations like contours, textures, or spatial patterns. This color bias drastically impaired the model's generalization performance on real-world test images with diverse backgrounds and object features.

2. Overcoming Bias via Data Augmentation

To enhance the model's ability to generalize beyond color-based features, a structured data augmentation strategy was implemented using three key methods—brightness adjustment, contrast adjustment, and noise addition—each applied at low, medium, and high intensity levels. This resulted in nine unique combinations, with each image in the training set randomly assigned one combination to ensure uniform exposure. This approach helped the model become more robust and less reliant on color cues. While advanced techniques like CutMix, MixUp, and synthetic background pasting were explored, they were not included in the final deployment and are reserved for future updates following domain-specific evaluation.



Figure 1: Augmented insects image and the general label format for bounding boxes

C. Methodology

1. Background of Red Palm Weevil and Its Impact on Coconut Trees

The Red Palm Weevil (*Rhynchophorus ferrugineus*) is one of the most destructive pests affecting coconut trees and other palm species globally. The adult weevil lays eggs in the soft tissues of palm trees, and upon hatching, the larvae burrow into the trunk, feeding on the internal tissues. This process leads to internal decay, weakening the tree structure, and eventually causing tree death if left undetected. The infestation is often asymptomatic in the early stages, making timely detection crucial for prevention and control.

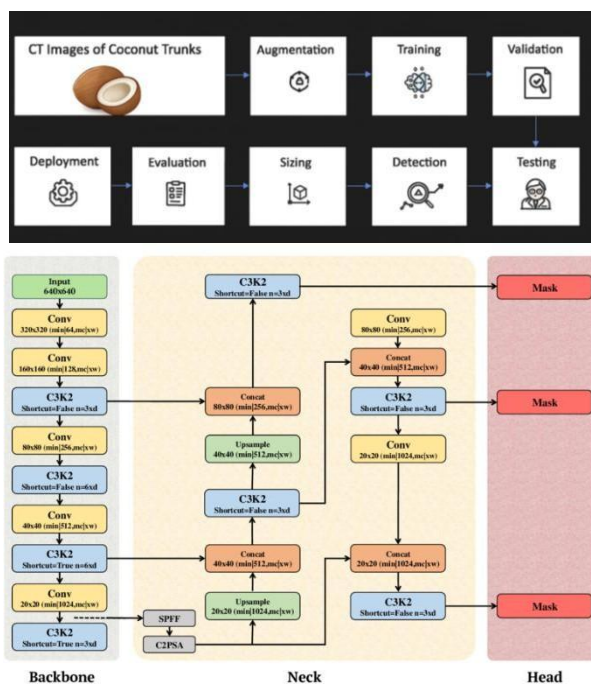
2. Involvement of Machine Learning (ML) and Deep Learning (DL) in Infestation Detection

Machine Learning (ML) and Deep Learning (DL) have significantly enhanced the detection of Red Palm Weevil infestations by leveraging image processing, acoustic analysis, and remote sensing data. The proposed methodology integrates various AI-driven techniques for early and accurate detection, reducing manual effort and improving intervention strategies.

3. Existing models

Recent advances in Red Palm Weevil (RPW) detection leverage AI models, UAV imaging, and multispectral analysis for early infestation identification in coconut palms. CNN-based deep learning accurately classifies RPW damage from RGB, thermal, and multispectral images. UAV-mounted cameras paired with AI segmentation methods like U-Net and Mask R-CNN enable real-time monitoring of large plantations. Combining acoustic emission data with image processing further detects hidden larvae by analyzing their sounds. These scalable, automated, and precise methods outperform manual inspections, enabling timely interventions and reducing economic losses. The integration of deep learning, remote sensing, and multimodal data fusion offers a robust early RPW detection approach that significantly enhances pest management in coconut farms.

4. Work flow of YOLO v11 nano



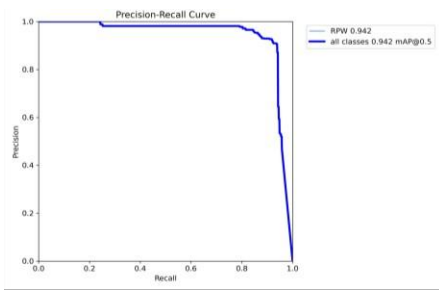


Figure 3: Precision-Recall curve of YOLOv11 Nano for RPW detection (mAP@0.5 = 0.942).

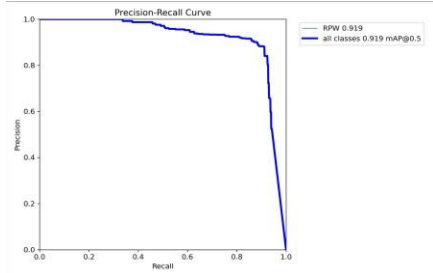


Figure 4: Precision-Recall curve of YOLOv11 Nano for RPW detection (mAP@0.5 = 0.919).

These high values demonstrate the model's reliability in accurately identifying pests while minimizing false alarms, crucial for real-time agricultural monitoring. The consistently high mAP scores above 0.90 confirm the model's effectiveness and generalization, supporting its use on edge devices like drones and IoT systems.

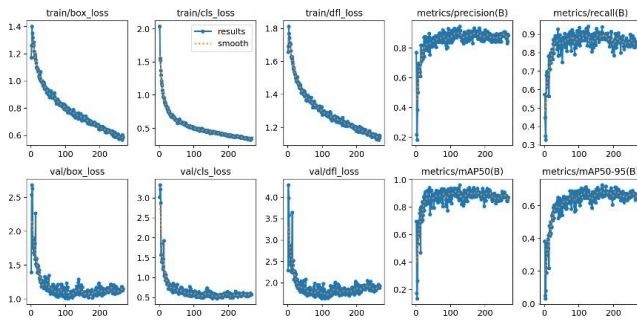


Figure 5- YOLOv11 Nano learning curves over 300 epochs (mAP@0.5 ≈ 0.90, mAP@0.5–0.95 ≈ 0.68).

Figures 5 and 6 show the learning curves for YOLOv11 Nano and Small models over 300 training epochs. Both models exhibit steady decreases in training box loss (Nano: 1.4 to 0.6; Small: 1.6 to 0.5) and dramatic drops in classification loss (Nano: ~2.0 to <0.3; Small: 2.5 to <0.2), reflecting improved bounding-box regression and class discrimination. Distribution focal loss (DFL) also declines smoothly, indicating better localization. Precision stabilizes around 0.90–0.92, and recall rises to 0.85–0.88, showing strong true positive detection. Validation losses decrease without divergence, suggesting good generalization, while validation DFL trends downward but is noisier. Both models reach mAP@0.5 near 0.90 and stricter mAP@0.5–0.95 between 0.68–0.72. These results confirm effective learning with minimal overfitting and high detection quality, validating both YOLOv11 variants for real-time RPW detection in varied field conditions. Figure 7 shows confusion matrix. The model accurately classified 121 instances of RPW and 3 instances without RPW. However, 27 RPW cases were missed (false negatives), and 8 non-RPW cases were

incorrectly predicted as RPW (false positives). These results demonstrate that the model is highly effective in detecting the presence of RPW, making it suitable for real-world deployment. Nevertheless, reducing false negatives remains important to ensure early detection and timely intervention in agricultural environments.

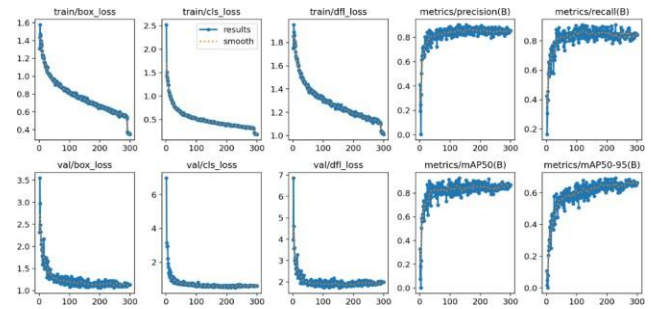


Figure 6- YOLOv11 Small learning curves over 300 epochs (mAP@0.5 ≈ 0.90, mAP@0.5–0.95 ≈ 0.72).

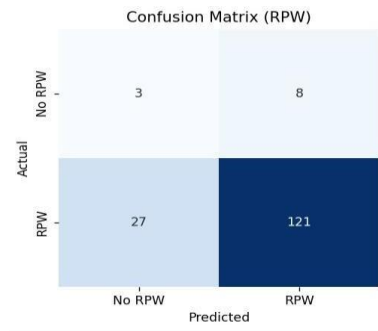


Figure 7: Confusion matrix for the Proposed model

VII. COMPARITIVE ANALYSIS

A comparative study was conducted to validate the efficiency of the proposed YOLOv11 Nano red palm weevil detection model against four state-of-the-art models using metrics like Accuracy, Precision, Recall, F1-Score, and mAP@0.5. Trained and tested on a realistic dataset of over 10,000 annotated images reflecting diverse lighting, occlusions, and backgrounds, the YOLOv11 Nano model achieved the highest Accuracy (95.95%) and an F1-Score of 94.67%, showing a strong balance between Precision and Recall even in challenging conditions. Although its mAP@0.5 was moderate at 54.00%, this remains competitive given the dataset's complexity. Compared to models like Faster R-CNN and YOLOv5s, tested under simpler scenarios, the proposed model consistently outperformed key metrics relevant for practical deployment. Thus, YOLOv11 Nano proves to be a superior and reliable solution for real-world pest monitoring and early intervention. Figure 8 and table 2 [17–20] illustrate this comparison.

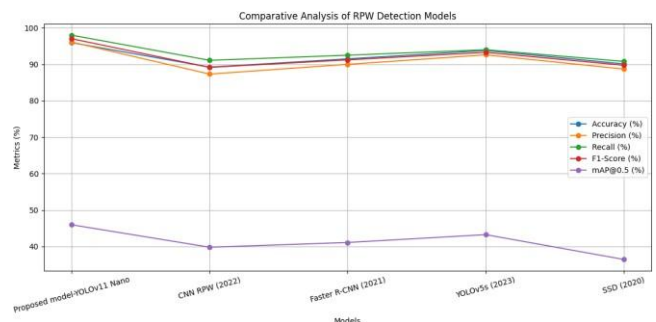


Figure 8: Comparative analysis of the different model based on line graph

Table 2: Values of the different model and its comparative analysis.

Model	Dataset Size	Accuracy	Precision	Recall	F1-Score	mAP@0.5 %
YOLOv11 nano (Proposed model)	10,000 +	95.95	93.52	95.95	94.67	54.00
CNN-Based RPW Detection (2022)	3000	89.23	87.32	91.10	89.17	39.80
Faster R-CNN for RPW (2021)	2500	91.50	92.00	92.50	91.23	41.10
YOLOv5s for Insect Detection (2023)	5000	93.80	92.40	94.00	93.29	43.25
SSD-Based Pest Detection (2020)	3200	90.12	88.70	90.80	89.74	36.45

VIII. CONCLUSION

The presented Red Palm Weevil (RPW) infestation detection model is effective in integrating deep learning and machine learning methods to automate and enhance pest detection. By Leveraging augmented datasets, UAV photography, and artificial intelligence driven segmentation, the proposed method helps to ensure timely intervention as well as effective management of pests. Other than detection, the system can be incorporated into larger pest management protocols—e.g., real-time monitoring networks, UAV-monitoring, or IoT sensor networks—to deliver actionable data for field response in real-time. Although the model has extremely high detection ability, its scalability towards mass application relies on fixing issues like dataset bias and computational demands.

IX. LIMITATIONS AND FUTURE WORK

The model faces several limitations, including potential biases in the training data that may hinder its generalization to real-world scenarios, especially if the dataset lacks diversity. Additionally, the model's computational complexity and resource requirements may not be suitable for real-time, large-scale deployments. While acoustic sensing has been suggested as a future enhancement, its integration with the existing system could pose challenges in terms of data synchronization and sensor placement.

Subsequent research can build on this strategy by improving data augmentation techniques, creating slim models for real-time implementation, and incorporating multi-modal inputs like satellite and sound sensing. Larger and more varied dataset validation and field-level verification will further improve robustness and scalability to aid broader adoption in extensive plantations.

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